

Dr Apurna Ghosh

Senior Lecturer | Undergraduate Course Coordinator

Department of Mining Engineering and Metallurgical Engineering

WA School of Mines: Minerals, Energy and Chemical Engineering

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PROFILE

- ✓ Performed the role of Acting Director of Graduate Research of WASM for 3 months
- ✓ Performed the role of Head of Discipline Mining Engineering for 1 month
- ✓ Chairperson of 14 PhD Thesis Committees
- ✓ Reviewed 3 PhD thesis from overseas Universities, Review 4-5 papers annually
- ✓ Organizing an international conference in collaboration with IIT Kharagpur on 'Geo-Resources Engineering, Energy, and Nature' (GREEN-2025), scheduled for 8-10 December 2025 in Kolkata, India
- ✓ Taken lead role in signing MoU between Curtin University and Coal India Limited (CIL)
- ✓ I have more than 25 years of combined experience in industry and academia, I am specialised in mine safety, and sustainability and was promoted to Senior Lecturer (ALC) in December 2019. The following are the summary of my key achievements since then.
- ✓ Designated as the Course Coordinator for BEng (Mining) and BSc (Mining) programs at Curtin.
- ✓ Played a lead role in renewing BSc (Mining) Curriculum and pivotal role in BEng (Mining) Curriculum.
- ✓ Supervised one PhD thesis on education topic with the award of Chancellor's Commendation for an exceptional Higher Degree by Research thesis.
- ✓ Delivered 5 invited presentations at international (1 Japan + 3 India + 1 China) forums.
- ✓ Completed 8 PhD (6 as Principal Supervisor and 2 as Associate Supervisor) supervisions and 1 MPhil Supervisions
- ✓ Published more than 50 peer-reviewed articles in journals and conference proceedings.
- ✓ Achieved an h-index 17 and i10-index 28 with 1332 citations (with the limited number of individuals working in this area of research, these indices are in fact among the highest, substantial and worthy of recognition).

Education

PhD in Mining Engineering

(2000 – 2003)

Indian Institute of Technology, Kharagpur

- An Epidemiological Investigation to Evaluate the Role of Personal and Impersonal Variates in Coal Miners Injuries.

Master of Mine Planning and Design

(1997-1999)

Indian School of Mines, Dhanbad

Bachelor of Mining Engineering

(1989-1993)

Indian Institute of Engineering Science and Technology, Shibpur

EMPLOYMENT

Senior Lecturer (Teaching & Research)

(2019 – present)

Lecturer (Teaching & Research)

(2016 – 2019)

Lecturer (Teaching focussed)

(2013 – 2016)

Curtin University

Assistant Professor

(2008 – 2013)

Senior Lecturer

(2005 – 2008)

Indian Institute of Engineering Science and Technology, Shibpur

Lecturer

(2003 – 2005)

National Institute of Technology, Rourkela, India

○ **Research**

PhD Supervisions Completed

1. Mitali Ghosh (2024). Doctor of Philosophy - Mining and Metallurgical Engineering on An Evaluation of Tertiary Educators' Perceptions of Online Teachnig Related Ergonomic Factors
2. Faiza Owais (2023). Doctor of Philosophy - Public Health on Risk Assessment as a Tool for Mobile Plant Operators for Sustainable Development: Lessons from the Western Australian Mining Industry
3. Zidong Zhao (2023). Doctor of Philosophy - Mining and Metallurgical Engineering on Evaluation of Surfactant Performance in Coal Dust Suppression for Underground Coal Mines
4. Yinping Chen (2021). Doctor of Philosophy - Mining and Metallurgical Engineering on Efficiency of Coal Dust Suppression by Surfactants in Underground Mines: Investigation of the Role of Various Factors
5. Joyce Sze Yee Chung (2021). Doctor of Philosophy - Mining and Metallurgical Engineering on A Mixed Integer Programming Approach for Transitioning from Open-Pit to Underground Mining
6. Michael Mayom Ajith (2020). Doctor of Philosophy - Mining and Metallurgical Engineering on A Mixed Method Investigation of Injuries Associated with Artisanal and Small-Scale Mining in Migori County, Kenya
7. Jinxin Huang (2020). Doctor of Philosophy - Mining and Metallurgical Engineering on Experimental and Numerical Study of Microwave Irradiation's Effect on Coal Petrophysical Property
8. Xingyun Guo (2019). Master of Philosophy (Mining and Metallurgical Engineering) on Assessing the Effectiveness of Eco-friendly Dust Suppressants Used to Abate Dust Emission from Mine Haul Roads
9. Shibani Chakraborti (2018). Doctor of Philosophy - Mining and Metallurgical Engineering on The Influence of Safety and Health Representatives in the Western Australian Mining Industries

Ongoing PhD Supervisions

10. Richard James Phelps. Master of Philosophy (Mining and Metallurgical Engineering) on Investigation of the Safety and Health Experiential Biases of Construction Industry Managers in the Western Australian Mining Sector
11. Shatrughan Thakur. Doctor of Philosophy - Mining and Metallurgical Engineering on Design and Development of an Energy-Efficient Hydraulic Power Pack Unit of a Chair Lift Man Riding System
12. Weifan Wu. Doctor of Philosophy - Mining and Metallurgical Engineering on Development and Evaluation of Eco-friendly Suppressant for Tailing Dust Suppression
13. Yoochan Kim. Master of Philosophy (Mining and Metallurgical Engineering) on Estimation of Iron Ore Price in reference to Major Economic Indices using Levenberg-Marquardt technique

PhD Chairperson

1. Bental Leandah Muvirimi. Master of Philosophy (Mining and Metallurgical Engineering) on Overbreak in Narrow-Vein Mining: Causes, Impacts, and Mitigation Strategies
2. Cristobal Ramiro Jeria Castaner. Master of Philosophy (Mining and Metallurgical Engineering) on Autonomous Haul Trucks Model to reduce costs and enhance Carbon Neutral Mining
3. Hyunjun Im. Doctor of Philosophy - Mining and Metallurgical Engineering on Development of Autonomous Overbreak Evaluation System using Overbreak Resistance Factor

4. Mark Atta-Danso. Doctor of Philosophy - Mining and Metallurgical Engineering on Understanding the Effect of High Shear Reactors on Complex Gold Ores Recovery
5. Nadia Kashif on Doctor of Philosophy - Mining and Metallurgical Engineering on Mineral Processing of Clay-Containing Ores
6. Sahar Agheli Goki. Master of Philosophy (Mining and Metallurgical Engineering) on Improvement of flotation performance of minerals containing lithium minerals
7. Sina Taheri. Doctor of Philosophy - Mining and Metallurgical Engineering on Uneven Break Prediction in Underground Mines Using Artificial Neural Network
8. Waheed Ahmed. Doctor of Philosophy - Mining and Metallurgical Engineering on Mineral processing techniques of gold ore
9. Walter Hocker. Doctor of Philosophy - Mining and Metallurgical Engineering on Controlling Galvanic interactions of sulphide minerals to improve flotation and leaching performance
10. Zhen Han. Doctor of Philosophy - Mining and Metallurgical Engineering on In cycle assessment of installed ground support capacity

Projects

- As Chief Investigator Project Title: Design and optimization of lithium and rare earth processing flowsheets, submitted to Coal India Limited \$6,000,000..
- As Chief Investigator Project Title: Application of Groundwater Model and Digital Elevation Model for storage of rainwater to Convert an Abandoned Open Pit Mine into an Asset sponsored by RTP, Department of Education, Government of Australia, \$70,000
- As Co-investigator Project Title: A comparison study on the dust suppressants for mine haul roads dust control sponsored by Waio Pty Ltd, \$46,300.00
- As Co-investigator Project Title: Fundamental understanding of nickel flotation selectivity by optimizing flotation chemistry sponsored by BHP Billiton Nickel West Pty Ltd, \$152,500.00
- Completed one project on “Risk Assessment of Occupational Injuries through Epidemiological Approaches in Underground Coal Mines” sponsored by All India Council of Technical Education (AICTE) on April 10, 2010 as chief Investigator.
- Ongoing project on ‘Role of Safety Leading Indicators and Individual Characteristics of Workers on Occupational Injuries in Coal Mines-A Machine Learning Approach’ for Scheme for Promotion of Academic and Research Collaboration (SPARC) in collaboration with National Institute of Technology, Surathkal (India).
- Recipient of Curtin eScholar Grant of \$10,000 for the project “Use mine simulations and visualisation models to design underground mine environments and safe management of hazards” in 2014 as Chief Investigator.

Teaching

- I have coordinated following courses:
- *Undergraduate*
 1. **Resource Estimation** (2017-till date). A 3rd year undergraduate MEA course. Effectively developed and implemented incorporating coding and softwares. MINE3010, class size: 60.
 2. **Hard Rock Mine Design** (2013 – 2015). 4th year undergraduate core MEA course. This is a project based project based unit run with industry supplied borehole data and exhaustive application of Mine Design softwares. MINE4002, class size: 110.
 3. **Mine Management** (2013 – till date). 4th year undergraduate core MEA course. Leader of the course team across the four MEA universities. Revised the course. Introduced new assessments and contents. MINE 4006, class size: 110.
 4. **Mining and Metallurgy** (2013- 2017). A second year undergraduate course taught in both

Kalgoorlie and Bentley campuses. MINE2001, class size: 50 (Bentley) + 20 (Kalgoorlie).

5. **Surface Mining Systems** (2019 - 2020). Project based 3rd year undergraduate core MEA course. Real-life simulated projects from both Coal and Metal mining sectors. . Tasked to coordinate in semester 2, 2019. MINE4005, class size: 62.
6. **Managing Mining Organisations** (2013 – 2014). It was a 12.5 credit course for Metallurgy 3rd year students. 3929, class size: 40.
7. **Rock Excavation** (2013 – 2014). Undergraduate course. Revised the course. Introduced new assessments and contents, class size: 40.

- **Postgraduate**

1. **Advanced resource Modelling and Estimation** (2017 – till date). A postgraduate course. Effectively developed and implemented incorporating real life problem, coding and softwares. Revised the course. Introduced new assessments and contents. MINE5011, class size: 35 (internal) +35 (fully online).
2. **Mine Design and Optimisation** (2013-till date). Project based postgraduate course deals with planning and design of both Coal and Metal Mines. MINE5010, class size: 18 (internal) + 20 (fully online).
3. **Mining Project Management and Operational Readiness** (2015 - till date). Postgraduate course. Revised the course. Introduced new assessments and contents. MINE6029, class size: 35 (internal) +35 (fully online).
4. **Ethics Safety and Regulations in Mining** (2015 -2016). Introduced and developed the postgraduate unit but stopped since 2017 due to restructuring. MINE6032, class size: 35 (internal) +35 (fully online).

I also supervise 7-10 students in each of the following undergraduate and postgraduate research units:

1. **ENGR6009 Professional Engineering Research Design Project 1**
2. **ENGR6010 Professional Engineering Research Design Project 2**

3. **ENGR4000 Engineering Industry Research Project 1**
4. **ENGR4001 Engineering Industry Research Project 2**

Credential Courses

I am involved in two credential courses: "Practical Geostatistics" and "Mine Closure." The first course, offered through the School of Earth and Planetary Sciences (EPS), is designed for geologists in the mining industry with little or no knowledge of geostatistics.

- **Services**

- I have been designated for the important role of **Undergraduate Course Coordinator** in WASM Mining Engineering Discipline.
- I am active **member of Undergraduate Curriculum Review Committee** to give valuable feedback to modernise the syllabus in line with rapid technological change in Mining Industry.
- In my role as the **Undergraduate Scholarship Assessor**; I have effectively managed a range of scholarships from individual industry and other sponsors.
- I actively **participate students' field trips**.

- I have been designated for the important role of **Inquiry Officer** and have completed investigations of 31 plagiarism cases as Inquiry Officer.
- I have served as **Health and Safety Representative** for building 703 for 5 years and submitted Work place Inspection reports for five labs/stores/workshops 20 times during the period.
- I am in the **Reviewers Committee** for the upcoming conference to be held in Perth during 2-4 December 2019 organised by Curtin University.
- I have acted as **International Advisory Committee Member** in International Conference on 'Global Challenges, Policy Framework and Sustainable Development for Mining of Mineral and Fossil Energy Resources' held at NITK, Surathkal, India during 17 -18 April, 2015.
- , I acted as an International Advisory Committee Member for the International Conference on 'Sustainable & Innovative Mining Practices' held at NIT Rourkela, India, from 16-18 November 2023.
- I served on the Reviewers' Committee for the International Conference on Mine Planning and Equipment Selection, held in Perth from 2-4 December 2019 by Curtin University.
- Acted as **Session Chair** in 7th International Conference on Mining Science and Technology held at CUMT Xuzhou, China during April 26-19, 2015.
- Regular **Reviewer** of number of peer reviewed reputable journals including Safety Science (Elsevier), Powder Technology (Elsevier), International Journal of Mining, Reclamation and Environment (Routledge, Taylor & Francis Group), International Journal of Environmental Research and Public Health (MDPI).
- I am also serving as First Aider and Fire Warden continuously.

Publications

Referred Journal Article

- 1) Phelps, R., J. Jansz, P. Chang, and A. Ghosh. 2025. "Corporate Manslaughter in the UK: Lessons for Australia." Social Sciences and Humanities Open 11
- 2) Roy, D. K., and A. K. Ghosh., "Transforming pollution sources into assets: Exploring innovative applications of abandoned open pit mines for rainwater storage and aquifer recharge." Journal of Hydrology 662 Inpress.
- 3) Phelps, R., J. Jansz, P. Chang, and A. Ghosh. 2024. "The Post-Harmonisation Health and Safety Challenges of Construction Industry Managers." Safety 10 (4)
- 4) Kumar, A., A. Senapati, A. Bhattacharjee, A. Ghosh, and N. Chau. 2024. "A practical framework to develop and prioritize safety interventions to improve underground coal miners' safety performance." Work (Reading, Mass.) 77 (2): 697-709.
- 5) Zhao, Z., P. Chang, A. Ghosh, G. Xu, Y. Liu, and R. Morla. 2024. "The influence of surfactants' ionicity on the performance of coal dust suppression." Advanced Powder Technology 35 (1)
- 6) Kim, Y., E. Topal, A. K. Ghosh, and M. W. A. Asad. 2024. "Investor Behavior in Gold, US Dollars and Cryptocurrency during Global Pandemics." Economies 12 (3)
- 7) Zhao, Z., P. Chang, G. Xu, A. Ghosh, and R. Morla. 2023. "Experimental study of surfactants' performance for suppressing coal dust with respirable size." Fuel 352
- 8) Kim, Y., A. Ghosh, E. Topal, and P. Chang. 2023. "Performance of different models in iron ore price prediction during the time of commodity price spike." Resources Policy 80
- 9) Zhao, Z., P. Chang, G. Xu, A. Ghosh, and R. Morla (2023). "Experimental study of surfactants' performance for suppressing coal dust with respirable size." Fuel 352.
- 10) Kim, Y., A. Ghosh, E. Topal, and P. Chang (2022). "Relationship of iron ore price with other major commodity prices." Mineral Economics 35 (2): 295-307.
- 11) Ghosh, M., Jansz, J. and Ghosh, A. (2022). "Effect of COVID-19 Pandemic on Traditional Teaching." International Journal on Studies in Education (IJonSE) 4(2), 107-129. (2): 107-129.
<https://doi.org/10.46328/ijonse.63>
- 12) Zhao, Z., Chang P., Xu G., Xie Q., and Ghosh A. (2022). "Comparison of static tests and dynamic tests

for coal dust surfactants evaluation: A review." *Fuel* 330.
<https://doi.org/10.1016/j.fuel.2022.125625>

- 13) Kim, Y., Ghosh A., Topal E. and Chang, P. (2022). "Relationship of iron ore price with other major commodity prices." *Mineral Economics* (<https://doi.org/10.1007/s13563-022-00301-x>)
- 14) Chang, P., Y. Chen, G. Xu, J. Huang, A. Ghosh, and W. V. Liu. 2021. "Numerical study of coal dust behaviors and experimental investigation on coal dust suppression efficiency of surfactant solution by using wind tunnel tests." *Energy Sources, Part A: Recovery, Utilization and Environmental Effects* 43 (17): 2173-2188.
- 15) Ajith M. M., and. Ghosh A. K., Jansz, J. (2021). A Mixed-Method Investigations of Work, Government and Social Factors Associated with Severe Injuries in Artisanal and Small-scale Mining (ASM) Operations, *Safety Science* 138, 105244 (<https://doi.org/10.1016/j.ssci.2021.105244>).
- 16) Chang, P., Xu, G., Chen, Y., **Ghosh, A.**, and Moridi, M. A. (2021). Improving coal powder wettability using electrolyte assisted surfactant solution. *Colloids and Surfaces A: Physicochemical and Engineering Aspects* 613 (<https://doi.org/10.1016/j.colsurfa.2020.126042>).
- 17) Zhao, Z., P. Chang, G. Xu, A. Ghosh, D. Li, and J. Huang. 2021. "Comparison of the coal dust suppression performance of surfactants using static test and dynamic test." *Journal of Cleaner Production* 328
- 18) Ajith, M., **Ghosh A.**, and Jansz J. (2020). Risk Factors for the Number of Sustained Injuries in Artisanal and Small-Scale Mining Operation. *Safety and Health at Work* (<https://doi.org/10.1016/j.shaw.2020.01.001>)
- 19) Zhu, S., Cheng J., Song W., Borowski M., Zhang Y., Yu B., Wang Y., Qi C., Tukkaraja P., Hua G., Xu Y., **Ghosh A.**, Wang C., Peng Y. (2020). Using seasonal temperature difference in underground surrounding rocks to cooling ventilation airflow: A conceptual model and simulation study. *Energy Science & Engineering*: 1-19. (<https://doi.org/10.1002/ese3.619>)
- 20) Chang, P., Zhao Z., Xu G., **Ghosh A.**, Huang J., and Yang T. (2020). Evaluation of the coal dust suppression efficiency of different surfactants: A factorial experiment. *Colloids and Surfaces A: Physicochemical and Engineering Aspects* 595. (DOI: 10.1016/j.colsurfa.2020.124686)
- 21) Chang P., Chen Y, Xu G., Huang J, **Ghosh A. K.**, and Liu W. V. (2019). Numerical study of coal dust behaviours and experimental investigation on coal dust suppression efficiency of surfactant solution by using wind tunnel tests, *Energy Sources, Part A: Recovery, Utilization, and Environmental Effects* (online), 1-16. doi:10.1080/15567036.2019.1639855.
- 22) Ajith M. M., and. **Ghosh A. K.**, (2019) "Economic and social challenges faced by injured artisanal and small-scale gold miners in Kenya." *SAFETY SCIENCE* 118: 841-852.
- 23) Ajith, M. M., and **Ghosh A. K.** (2019). "Comparison of parameters for likelihood and severities of injuries in artisanal and small-scale mining (ASM)." *SAFETY SCIENCE* 118: 212-220.
- 24) Cheng, J., Zhang, X. and **Ghosh, A. K.** (2017). "Explosion risk assessment model for underground mine atmosphere." *Journal of Fire Sciences* 35 (1): 21-35.
- 25) Cheng, J., Li, S., Zhang, F., Zhao, C., Yang, S. and **Ghosh, A. K.** (2016). "CFD modelling of ventilation optimization for improving mine safety in longwall working faces." *Journal of Loss Prevention in the Process Industries* 40: 285-297.
- 26) Chung, J., Topal, E., and **Ghosh, A. K.** (2016). "Where to make the transition from open-pit to underground? Using integer programming." *Journal of the Southern African Institute of Mining and Metallurgy* 116 (8): 801-808.
- 27) **Ghosh, A. K.** and Wang, S. (2014). "Evolution of underground coal mine explosion law in Australia, 1887-2007." *Journal of Australasian Mining History*, Volume 12, pp. 81-97.
- 28) **Ghosh, A. K.** (2014). Improving coal recovery from longwall top coal caving. *Journal of Mines Metals and Fuels*, Volume 62(3), pp. 51-64.
- 29) **Ghosh, A. K.** and Garg, V. (2013): Derivation of Morphometric Parameters Using Cartosat-1 Data, *Journal of Water and Land Use Management*, Volume 13 (1), pp. 63-68.
- 30) Biswas, R. and **Ghosh, A. K.** (2012): Effect of blast design parameters on fragmentation - an application of Kuz-Ram Model, *Journal of Mines Metals and Fuels*, Volume 60, No. 9&10, pp. 206-215.
- 31) **Ghosh, A. K.** (2010): Risk Assessment of Occupational Injuries in Underground Coal Mines, *Journal of Mines, Metals & Fuels*, Volume 58, No. 9, pp. 243-248.
- 32) **Ghosh, A. K.** and Bhattacharjee, A. (2009): Loglinear Model for Assessment of Risk Factors of

Occupational Injuries in Underground Coal Mines, *Journal of Geology and Mining Research*, Volume 1(2), pp. 25-33.

- 33) **Ghosh, A. K.** and Bhattacharjee, A. (2007): Predictors of Occupational Injuries among Coal Miners: A Causal Analysis, *Mining Technology (Trans. Inst. of Materials, Minerals and Mining, UK and Australian Instt. of Mining and Metallurgy)*, Volume 116.1, pp. 16-24.
- 34) Bhattacharjee, A.; Bertand, J. P.; Meyer, J. P., **Ghosh, A. K.**, Chau, N.; et al. (2007): Relationships between Physical Job Tasks and Living Conditions with Occupational Injuries in Coal Miners, *Industrial health*, Volume 45, pp. 352-358.
- 35) **Ghosh, A. K.**; Bhattacharjee, A. and Chau, N. (2004): Relationships of Working Conditions and Individual Characteristics to Occupational Injuries: A Case-control Study of Coal Miners, *Journal of Occupational Health*, Volume 46, pp. 470-478.
- 36) Bhattacharjee, A.; Chau, N.; **Ghosh, A. K.** and Benamghar, L et al. (2003): Relationships of Job and Some Individual Characteristics with Occupational Injuries in Employed People: A Community-Based Study, *Journal of Occupational Health*, Volume 45, pp. 382-391.
- 37) **Ghosh, A. K.** and Bhattacharjee, A. (2003): Role of Individual Characteristics of Workers in Mine Accidents: A Case-Control Study, *Minetech*, Volume 24(1), pp. 43-48.

Referred Conference Publications

- 1) Dube A., Ghosh A. K. (2023). Application of Groundwater Modelling and Digital Elevation Model for Determination of the Catchment Area to Convert an Abandoned Open Pit Mine into an Asset. In 6th International Conference on Opencast Mining Technology & Sustainability ICOMS 6.0 22-23 December 2023, Singrauli, MP, India
- 2) Gautam, R.K., Ghosh, A.K., Chang, P. (2022). Study of the Effectiveness of Hydrogen Vehicles in Mining Industry. In the proceedings of International Conference On Sustainable Mining Options... Way Ahead, 3-5 June 2022, Nagpur, India
- 3) Alston, N., Chang, P., Zhao, Z. and **Ghosh A.K.** (2021). Laboratory and Field Testing of Surfactants Used to Meet New Workplace Exposure Standards for Respirable Dust. In Coal Mines Proceedings of the 2021 Resource Operators Conference, University of Southern Queensland, QLD, February 2021, pp. 257-265.
- 4) **Ghosh A. K.** (2016). Application of Shuttle Radar Topography Mission Data in Hydrological Analyses of Super Pit and Griffin Coalmine. In Ninth AusIMM Open Pit Operators' Conference held at Kalgoorlie, Australia, November 15-16, 2016, pp. 162-169.
- 5) Chung J, Asad, M., Topal, E., **Ghosh, A. K.** (2016). Determination of the Transition Point from Open Pit to Underground Mining. In Ninth AusIMM Open Pit Operators' Conference, November 15-16, 2016, pp. 96-103.
- 6) **Ghosh, A. K.** (2015): Lost Work Days as a Measure of Risk Assessment in Underground Coal Mines. In: *The 7th International Conference on Mining Science and Technology* held at Xuzhou, China during April 26-19, 2015, pp. 155-161.
- 7) **Ghosh, A. K.** (2012): Effect of Surface Mining on Groundwater Table. In: *National Seminar on Development of Coal and Mineral Resources – Economic, Technological and Environmental Issues* held at Kolkata on February 23 – 24, pp. 30.
- 8) **Ghosh, A. K.** (2008): Application of Artificial Neural Network in Roof Characteristic Determination. In: *Proceedings of National Seminar on Rock Machine Interaction in Excavations*, Varanasi, March 7 – 8, pp. 173-182.
- 9) Chau, N.; Benamghar; L.; Mur, J. M.; Bhattacharjee, A.; **Ghosh, A. K.**; Bertand, J. P., Meyer, J. P.; Guillemin, F. and Ravaud, J. F. (2005): Associations of Occupational Hazards and Individual Characteristics with Occupational Injuries and Disabilities in Lorraine Coal Miners. In: *Proceedings of the International Symposium on Advances in Mining Technology and Management*, Kharagpur, November 30-December 2, pp. 543-552.
- 10) Bhattacharjee, A.; Chau, N. and **Ghosh, A. K.** et al. (2004): Associations of Safety Environment, Training, and Individual Characteristics with Occupational Injuries: A Case-Control Study, In: *Proceedings of 17th International Symposium on Epidemiology in Occupational Health*, Melbourne, Australia, October 13-15, pp. 222-227
- 11) Bhattacharjee A., Chau N., Benamghar L., Michaely J.P., **Ghosh, A. K.**, Guillemin F., Ravaud J.F., Mur J.M., and Lorhandicap group. (2002): Associations between individual characteristics and

occupational injuries: a community-based study. In: *16th EPICOH 2002, Barcelona, 11-13 September 2002*.

- 12) **Ghosh, A. K.** and Bhattacharjee, A. (2002): The Role of System Factors including Personal Factors in Coal Mine Accidents – An Epidemiological Study, In: *Proceedings of the International Conference on Mineral Industry: Issues of Economics, Environment and Technology*, September 18-20, 2002, MGMI, Kolkata, India, pp.132-139.
- 13) **Ghosh, A. K.**; Bhattacharjee, A.; Ray, S.C. and Sharan, M. B. (2001): Behavioural Safety in the Wake of Mechanisation and Automation of Mineral Industry, *Proceedings of the International Conference on Mechanisation & Automation-the Future for Mineral Industry*, July 2001, MGMI, Kolkata, pp.171-177.
- 14) Bhattacharjee, A.; Ray, S. C.; Mukhopadhyay, S.K.; Sharan, M. B. and **Ghosh, A.K.** (2000): An Investigation into Personal and Impersonal Variates in Coal Mine Accidents, In: *Proceedings of the International Conference on Technology Management for Mining, Processing and Environment (TMMPE)*, December 1-3, 2000, IIT, Kharagpur, pp. 297-302.
- 15) **Ghosh, A. K.** and Murthy, V. M. S. R. (1999): Higher Pull in Solid Blasting –Problems and Remedies, In: *Proceedings of National Conference on Drill-Blast’99*, January 17-18, 2000, Bhubaneswar, pp. 79-84.
- 16) **Ghosh, A. K.** and Murthy, V. M. S. R. (1999): Design Aspects of Solid Blasting in Coal Development Headings - Some Safety and Productivity Considerations, In: *Proceedings of National Conference on Practices and Prospects of Mining Industry in Central India* held at RKN college of Engineering, Nagpur on January 2, 1999, pp. 63-69.

Book Chapters (Research)

- 1) Zhao, Z., A. Ghosh, P. Chang, and Y. Liu. 2023. "Evaluation of different surfactants' performance in varying coal dust concentration through logistic regression analysis." In *Underground Ventilation*. 313-321. 2023

Invited Speaker

1. **Ghosh A.** (2024) "Safety Data Analytics Applications in Mining and Other Core Industries" at NITK Surathkal, India, on 6-12 March 2024.
2. **Ghosh A.** (2021). "Safety Research in the Wake of Mine Automation" at Hokkaido University, Japan, on 28 September 2021,
3. **Ghosh A.** (2020). Presented speech on Planning & Design aspects in Transition from open-pit to underground in International Webinar On Frontiers in Mining & Geosciences with Special focus on Remote Sensing, Rock Mechanics & Mine Planning organised by Department of Mining Engineering, National Institute of Technology, Rourkela, India on 8 Nov 2020.
4. **Ghosh A.** (2019). North China University of Science and Technology has invited to deliver lecture on Mine Safety and visit research labs and teaching facilities and meet with academics for collaborative research during 9-15 December 2019.
5. **Ghosh A.** (2016). Invited by China University of Mining & Technology, Beijing, China during December 27 - 31, 2016 to deliver lecture on 'Recent Trends on Mine Safety' and collaboration with Academic Staff.
6. **Ghosh A.** (2016). M/s BHP Billiton invited to deliver lecture on 'Incorporating the Sophistication of Human Thinking within the Complex Computation Power of Machines' in Automation Week during 8-10 August 2017 at Perth.

Industry Engagement

- Invited Mr David Muller of DMIRS for delivering lecture on WA Mines Act and Regulations (
- Invited Mr Jeremy Peters of Burnt Shirt to deliver lecture on Joint Ore Reserve Committee (JORC) code
- Invited Mr Uchchhal Ray of BHP to deliver lecture on Mine Project Management

- Invited Murray Rayner, Principal Geologist - Resource Evaluation, Rio Tinto Copper Technical to deliver lecture on Resource Estimation

Professional Memberships

- Active member of the Australasian Institute of Mining and Metallurgy (AusIMM).
- Life member of the Mining, Geological, Metallurgical Institute of India (MGMI)